

Special integrating factors for exact ODE

Integrating Factor

Definition 3. If the equation

$$(1) \quad M(x, y)dx + N(x, y)dy = 0$$

is not exact, but the equation

$$(2) \quad \mu(x, y)M(x, y)dx + \mu(x, y)N(x, y)dy = 0,$$

which results from multiplying equation (1) by the function $\mu(x, y)$, is exact, then $\mu(x, y)$ is called an **integrating factor**[†] of the equation (1).

Example 1 Show that $\mu(x, y) = xy^2$ is an integrating factor for

$$(3) \quad (2y - 6x)dx + (3x - 4x^2y^{-1})dy = 0.$$

$$\frac{\partial}{\partial y} \downarrow = 2 \neq \frac{\partial}{\partial x} \downarrow = 3 - 8xy^{-1} \Rightarrow \text{not exact}$$

$$\begin{aligned} (2xy^3 - 6x^2y^2)dx + (3x^2y^2 - 4x^3y)dy &= 0 \\ \frac{\partial}{\partial y} \downarrow &= 6xy^2 - 12x^2y \\ \frac{\partial}{\partial x} \downarrow &= 6xy^2 - 12x^2y \Rightarrow \text{exact} \end{aligned}$$

How to find $\mu(x, y)$?

$$\mu(x, y) \text{ s.t. } \frac{\partial}{\partial y} (\mu(x, y) M(x, y)) = \frac{\partial}{\partial x} (\mu(x, y) N(x, y))$$

$$\frac{\partial \mu}{\partial y} \cdot M + \mu \frac{\partial M}{\partial y} = \frac{\partial \mu}{\partial x} N + \mu \frac{\partial N}{\partial x} \quad \leftarrow \text{PDE hard}$$

Ass 1: $\mu(x)$: $\frac{\partial \mu}{\partial y} = 0 \Rightarrow \mu \frac{\partial M}{\partial y} = \mu' N + \mu \frac{\partial N}{\partial x}$

$$\frac{d\mu}{dx} N = \mu \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) \leftarrow \text{separable} \Rightarrow \frac{d\mu}{\mu} = \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} dx$$

if $\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) / N$ depends on x only, then

$$\ln \mu = \int \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} / N \right) dx$$

Ass 2: $\mu(y)$: ...

$$\boxed{\mu(x) = e^{\int \frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N} dx}}$$
$$\boxed{\mu(y) = e^{\int \frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M} dy}}$$

Special Integrating Factors

Theorem 3. If $(\partial M/\partial y - \partial N/\partial x)/N$ is continuous and depends only on x , then

$$(8) \quad \mu(x) = \exp \left[\int \left(\frac{\partial M/\partial y - \partial N/\partial x}{N} \right) dx \right]$$

is an integrating factor for equation (1).

If $(\partial N/\partial x - \partial M/\partial y)/M$ is continuous and depends only on y , then

$$(9) \quad \mu(y) = \exp \left[\int \left(\frac{\partial N/\partial x - \partial M/\partial y}{M} \right) dy \right]$$

is an integrating factor for equation (1).

Method for Finding Special Integrating Factors

If $M dx + N dy = 0$ is neither separable nor linear, compute $\partial M/\partial y$ and $\partial N/\partial x$. If $\partial M/\partial y = \partial N/\partial x$, then the equation is exact. If it is not exact, consider

$$(10) \quad \frac{\partial M/\partial y - \partial N/\partial x}{N}.$$

If (10) is a function of just x , then an integrating factor is given by formula (8). If not, consider

$$(11) \quad \frac{\partial N/\partial x - \partial M/\partial y}{M}.$$

If (11) is a function of just y , then an integrating factor is given by formula (9).

Example 2 Solve

(12) $(2x^2 + y)dx + (x^2y - x)dy = 0$.

$$\frac{\partial}{\partial y}(2x^2 + y) = 1 \quad \text{not exact}$$
$$\frac{\partial}{\partial x}(x^2y - x) = 2xy - 1$$

$$1 - (2xy - 1) = -2xy + 2 = 2(1 - xy)$$

$$\frac{2(1 - xy)}{2x^2 + y} \leftarrow \begin{array}{l} \text{not dependent only on } x \text{ or } y \\ \text{no special int factor} \end{array}$$

$$\frac{2(1 - xy)}{x^2y - x} = \frac{2(1 - xy)}{-x(1 - xy)} = \frac{2}{-x} \Rightarrow \mu(x) = e^{\int \frac{2}{-x} dx} = x^{-2}$$

$$* x^{-2} \Rightarrow (2 + x^{-2}y)dx + (y - x^{-1})dy = 0$$

$$\frac{\partial}{\partial y} \downarrow = x^{-2} = \frac{\partial}{\partial x} \downarrow = x^{-2} \Rightarrow \text{exact}$$